

Introduction

Ice cover in the Great Lakes highly variable and has significant influence on regional climate, ecology, and economy. Spatially resolved ice cover ($\Delta x = 1.25\text{KM}^2$) has been recorded since 1973 and is available as daily files throughout each ice season¹. Previous work has assessed large-scale (i.e. lake or basin-wide) ice cover fluctuations through time and their relationship to global weather phenomena^{2,3,4,5,6}. We follow Mason et al. (2016)² and use the Great Lakes Aquatic Habitat Framework (GLAHF)⁷ to analyze ice cover on a sub-regional basis for the below metrics and analyze how morphometry and atmospheric conditions impact ice cover in the Great Lakes.

- **Annual Max Ice Cover (AMIC):** maximum percent during a given season
- **Seasonal Average Ice Cover (JFM):** mean daily ice cover: Jan 1 - Mar 31
- **Ice Duration:** Total (non-consecutive) days[†] where ice cover exceeded 10%

Methods

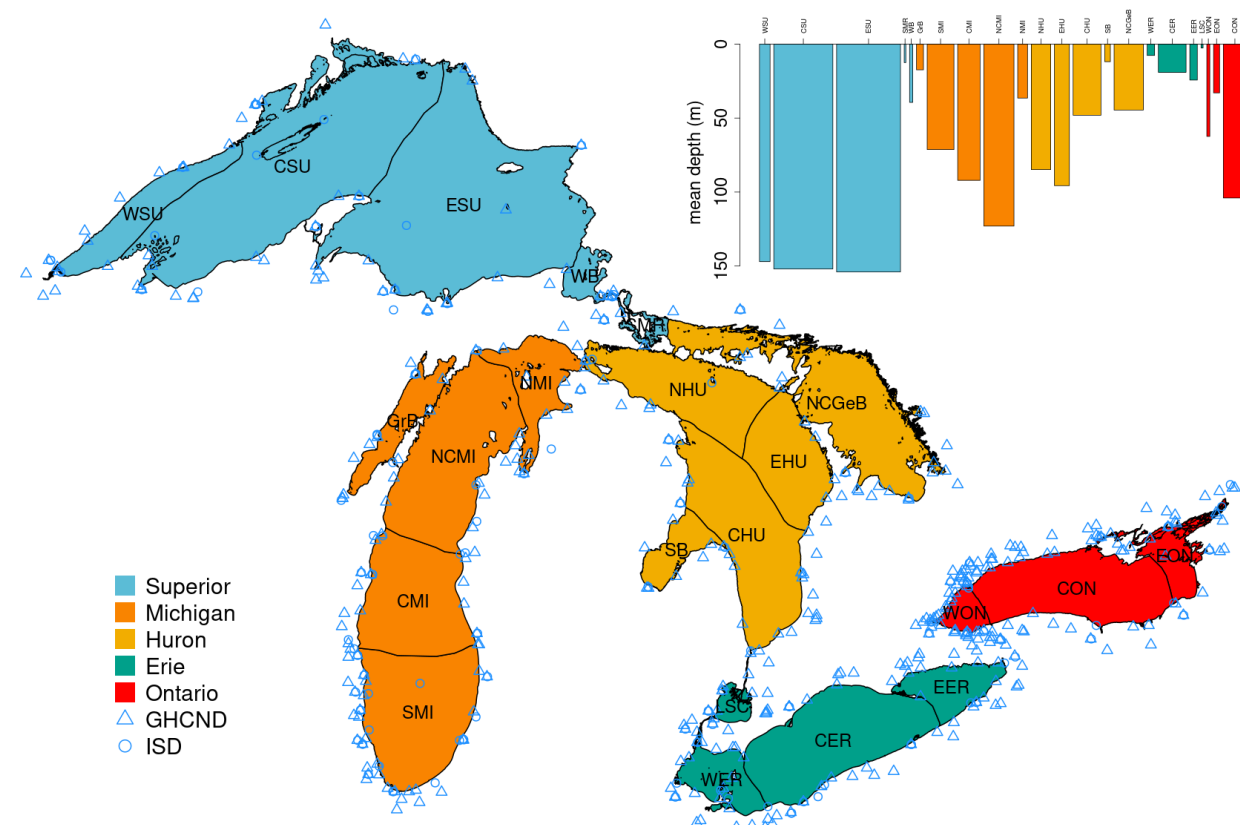


Figure 1. Spatial delineation of GLAHF subregions and selected met station locations (GHCND⁸, ISD⁹). The upper right figure shows depth and (relative) surface area of each subregion in the y and x directions, respectively.

1. Do ice trends exist at these spatial scales and how are they related to lake morphometry?
2. How do atmospheric conditions impact ice formation?

To address #1, the three metrics are calculated for each GLAHF subregion and ice season and then Mann-Kendall monotonic trend test¹⁰ is applied to check for statistical significance. Additionally correlations between trend magnitude and morphometrics (depth, area, latitude) are identified. To address #2, station data^{8,9} is aggregated for each subregion based on a 20 KM radius and correlations between ice metrics and monthly mean meteorological surface variables are identified.

†: days are only counted Jan 1 - Mar 31 due to shorter ice reporting windows during 1970's and 80's.

Analysis

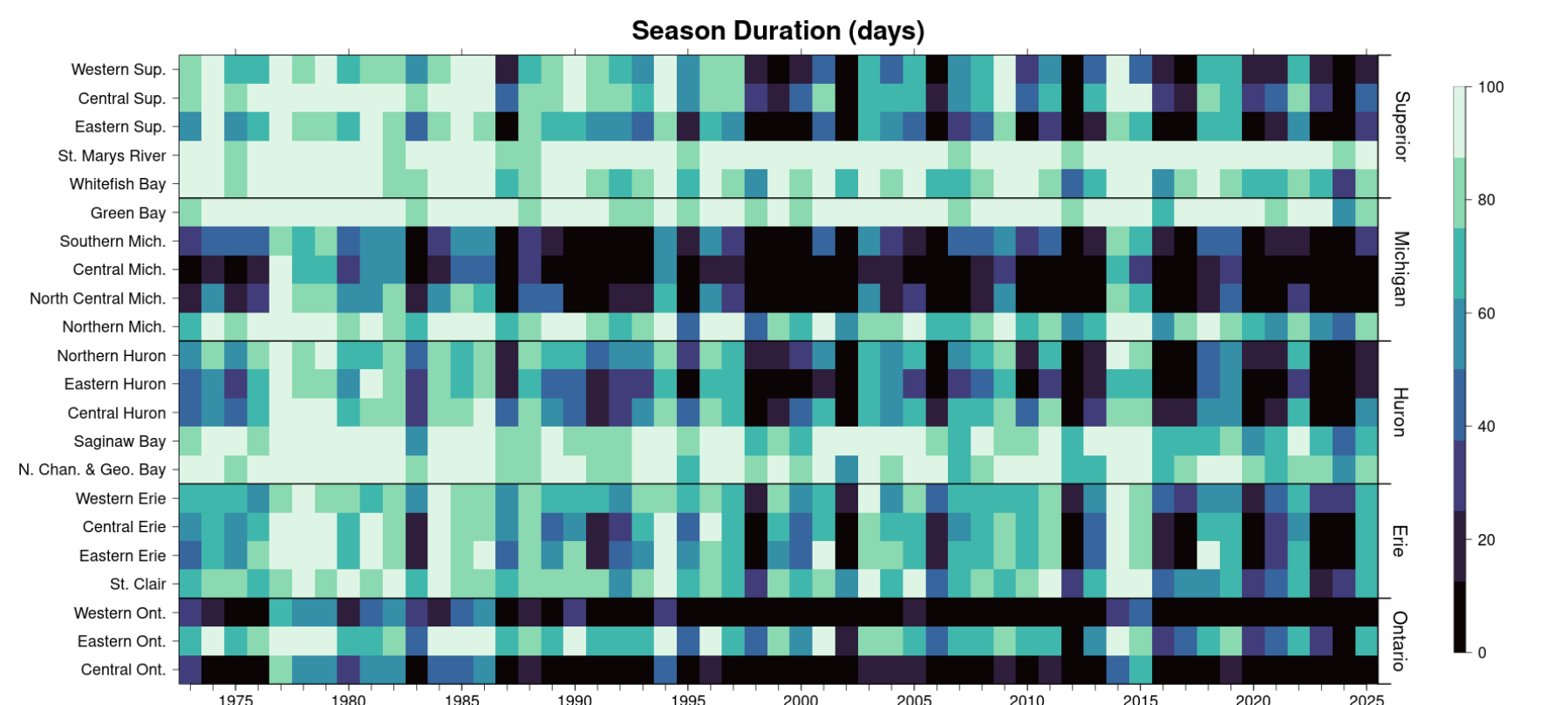


Figure 2. Evolution of one of three metrics analyzed (ice duration) for each subregion.

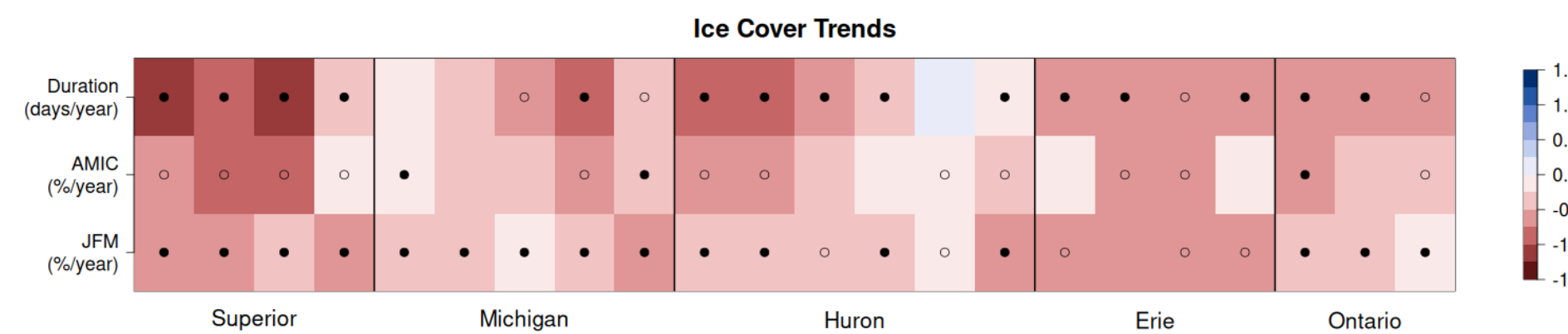


Figure 3. Linear trends for each metric. Open (solid) dots indicate 95% (99%) statistical significance. Columns correspond to rows in fig 2.

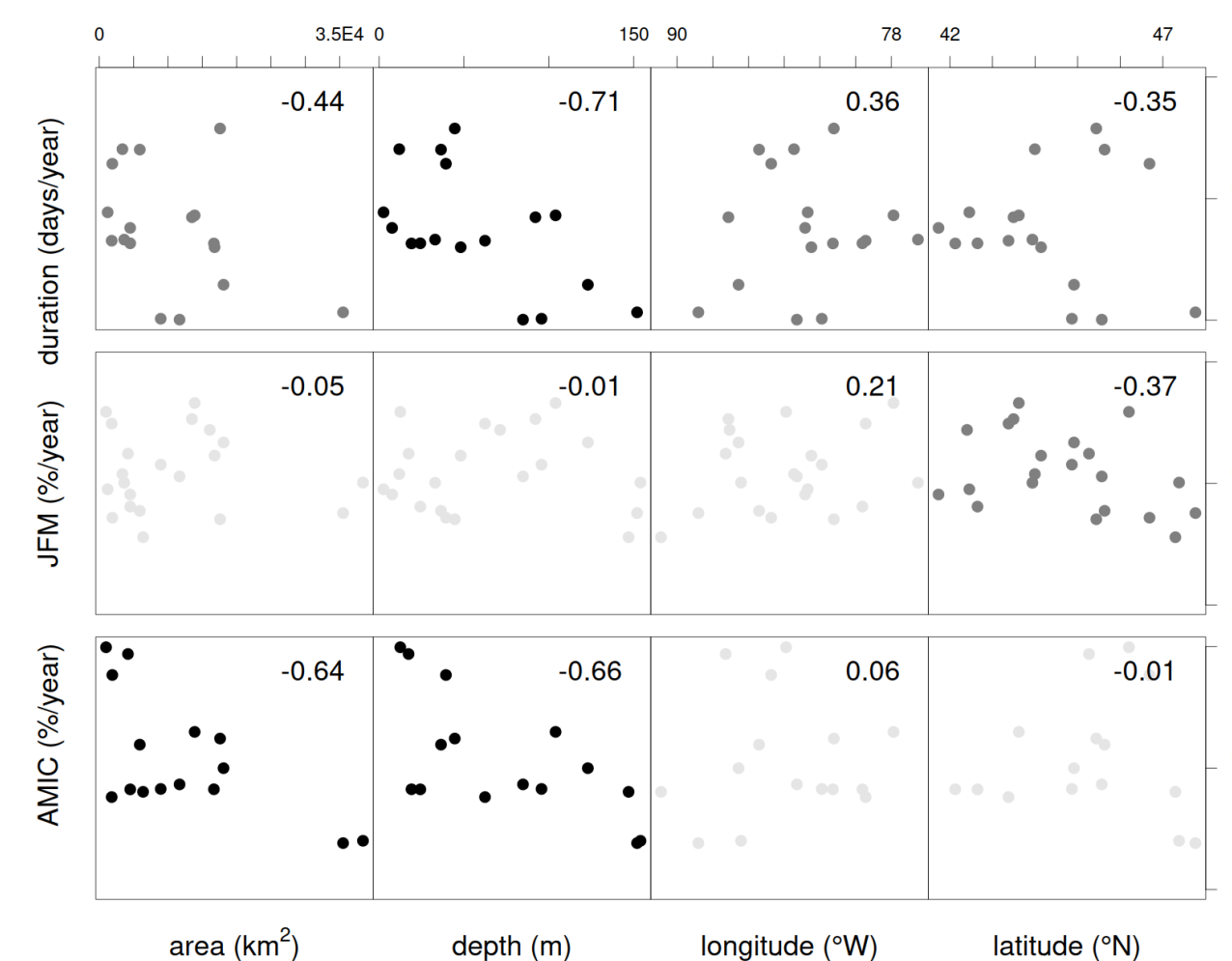


Figure 4. Correlations between statistically significant ice metric trends (vertical axes) and morphometrics (horizontal axes). Moderate correlations (pearson's $r > 0.3$) are shown in dark grey and strong correlations (> 0.6) are black.

Season duration decrease is visible in some regions but not apparent in others (see fig. 2). Regions that show decrease are often offshore and deep: Superior (WSU, CSU, ESU), Huron (NHU, EHU, CHU), and Ontario (WON, CON). This may be due to warmer subsurface temps^{11,12} during later years providing heat to surface waters and preventing freezing, whereas shallow regions may be able to consistently release all heat from the water column.

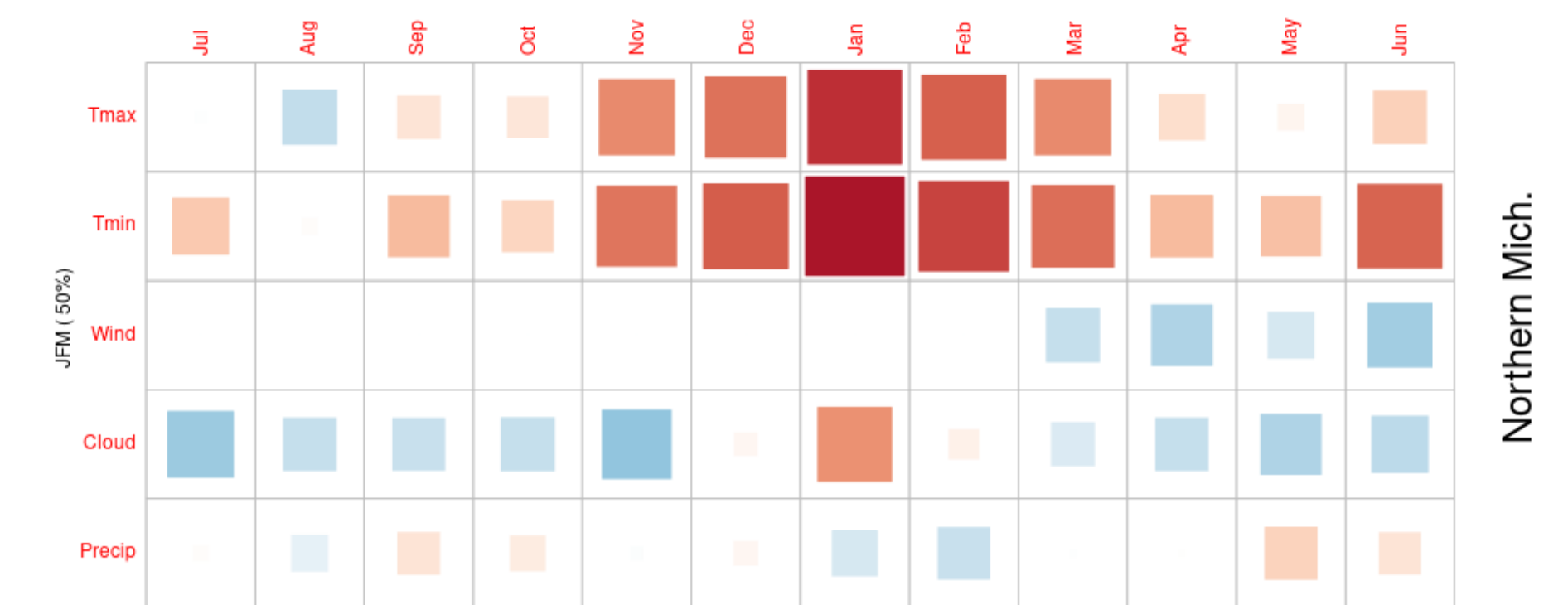


Figure 5. Correlation between Seasonal Ice Cover (JFM) and local meteorology. Meteorological variables are aggregated by month (columns) and state variable (rows). Larger squares indicate stronger correlations and blue (red) indicates direct (indirect) correlation.

As expected, ice cover showed strong inverse correlation with air temperature. Wind speed showed moderate correlation to ice in some regions which may be due to wind's influence on heat distribution in the water column via mixing (though this impact is more relevant during fall overturn). The analyses presented here are a small fraction of possible correlation pairs and more work is needed, including different time periods and other metrics like ice onset/offset.

Summary

- A majority of subregions show significant decreasing trends in all three metrics
- Seasonal average trends were significant in ALL 22 subregions
- Max Ice Cover showed the least significant trends (17/22 regions)
- Air temperature was strongly anti-correlated with ice cover for all metrics
- More work needed: comprehensive assessment of ice cover and meteorological data correlations

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